BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

CE-403: Database Administration (PostgreSQL)

Elchin Salahzada

Lecture 2

CE-403: Database Management Systems
Baku Higher Oil School : Computer Engineering

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

SQL Fundamentals I

- SQL Syntax & Rules
- Data Definition Language (DDL)
- Data Types in SQL
- Data Manipulation Language (DML)
- The SELECT Statement Basics
- Filtering Data with WHERE
- Sorting Data with ORDER BY
- Operators (comparison, logical, BETWEEN, IN, LIKE)
- Lab Activity Overview

Learning objectives

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Write basic SQL syntax correctly, following standard formatting and rules
- Create, alter, and drop database objects (tables) using DDL commands
- Identify and apply appropriate SQL data types when defining table columns
- Insert, update, and delete records in a table using DML commands
- Retrieve data from a table using the SELECT statement, and filter and sort results using WHERE and ORDER BY
- Apply comparison and logical operators (BETWEEN, IN, LIKE) to write more precise queries

SQL Syntax & Rules

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- SQL (Structured Query Language) is used to communicate with relational databases
- SQL statements end with a semicolon (;)
- Keywords (e.g., SELECT, FROM, WHERE) are not case-sensitive, but convention is to write them in UPPERCASE
- Identifiers (table/column names) are case-sensitive in some DBMS based on usage of “” (e.g., PostgreSQL)
- Statements can span multiple lines for readability
- Comments:
  - Single-line: -- comment
  - Multi-line: /* comment */
- String values are enclosed in single quotes ('value')
- Basic structure of a query:
  - SELECT column_name FROM table_name WHERE condition;

SQL Syntax & Rules

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- PostgreSQL's CLI tool (psql) has special backslash commands to inspect the database
- \l – List all databases
- \c dbname – Connect to a specific database
- \dt – List all tables in the current database
- \d table_name – Describe a table's structure (columns, types, keys)
- \du – List all users/roles
- \dn – List all schemas
- \q – Quit psql
- These are client-side commands, not SQL — they only work inside psql, not in general SQL scripts

Data Definition Language (DDL)

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- DDL commands define and modify database structure
  - CREATE – Creates a new database object
    - CREATE TABLE students (
    - student_id SERIAL PRIMARY KEY,name VARCHAR(50));
  - ALTER – Modifies an existing table structure
    - ALTER TABLE students ADD COLUMN email VARCHAR(100);
  - DROP – Permanently deletes a table or database
    - DROP TABLE students;
  - TRUNCATE – Removes all rows from a table, but keeps its structure
    - TRUNCATE TABLE students;
- PostgreSQL uses SERIAL for auto-incrementing primary keys (instead of AUTO_INCREMENT in MySQL)

Data Types

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

Category	Data Type	Description	Example Use
Numeric	`INTEGER`, `BIGINT`	Whole numbers of varying sizes	Unique IDs, counts
	`NUMERIC`, `DECIMAL`	Fixed-point numbers with precision	Balances, financial data
Character	`VARCHAR`, `TEXT`	Variable-length text	Names, descriptions
Date/Time	`DATE`, `TIMESTAMP`	Dates and timestamps	Event dates, logs
Boolean	`BOOLEAN`	True/False values	Status flags
Array	`ARRAY`	List of values of the same type	List of tags, multiple emails
Special	`UUID`, `JSON`	Unique identifiers, JSON data	User IDs, product attributes
Geometric	`POINT`, `LINE`	Geometric data types for coordinates	Location data

Data Types

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Numeric Types
  - INTEGER / INT – whole numbers
  - SERIAL – auto-incrementing integer (used for IDs)
  - NUMERIC(p,s) / DECIMAL – exact precision (e.g., currency)
  - REAL, DOUBLE PRECISION – floating-point numbers
- Character/String Types
  - VARCHAR(n) – variable-length string with limit
  - CHAR(n) – fixed-length string
  - TEXT – unlimited length string
- Date/Time Types
  - DATE – calendar date
  - TIME – time of day
  - TIMESTAMP – date and time combined
  - INTERVAL – span of time
- Boolean Type
  - BOOLEAN – TRUE / FALSE / NULL
- Other Useful Types
  - UUID – universally unique identifier
  - JSON / JSONB – store JSON data (JSONB is binary, faster to query)
  - ARRAY – store arrays of values

Data Manipulation Language

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- DML commands manage the data stored inside database objects
  - SELECT – Retrieves data from a table
    - SELECT * FROM students;
    - SELECT name, email FROM students WHERE student_id = 1;
  - INSERT – Adds new rows into a table
    - INSERT INTO students (name, email) VALUES ('Asena', 'asena@example.com');
  - UPDATE – Modifies existing rows in a table
    - UPDATE students SET email = 'new@example.com' WHERE student_id = 1;
  - DELETE – Removes specific rows from a table
    - DELETE FROM students WHERE student_id = 1;
- Unlike DDL commands, DML changes can be rolled back with ROLLBACK if run inside a transaction (BEGIN ... COMMIT)

The SELECT Statement

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- SELECT retrieves data from one or more tables
  - SELECT column1, column2
  - FROM table_name;
- Select all columns using *
  - SELECT * FROM students;
- Select specific columns
  - SELECT name, age FROM students;
- Column aliasing with AS (renames output column)
  - SELECT name AS student_name FROM students;
- DISTINCT – removes duplicate rows from results
  - SELECT DISTINCT age FROM students;
- LIMIT – restricts number of rows returned (PostgreSQL-specific)
  - SELECT * FROM students LIMIT 5;

Filtering Data with WHERE

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- WHERE filters rows based on a specified condition
  - SELECT column1, column2
  - FROM table_name
  - WHERE condition;
- Comparison operators: =, <> (or !=), <, >, <=, >=
  - SELECT * FROM students WHERE age > 20;
- Logical operators: AND, OR, NOT
  - SELECT * FROM students WHERE age > 20 AND name = 'Alice';
- NULL checks – use IS NULL / IS NOT NULL (never = NULL)
  - SELECT * FROM students WHERE email IS NULL;
- WHERE is evaluated before SELECT — filters raw table rows, not aliased columns

Sorting Data with ORDER BY

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- ORDER BY sorts query results based on one or more columns
  - SELECT column1, column2
  - FROM table_name
  - ORDER BY column1;
- Sort direction:
  - ASC – ascending order (default)
  - DESC – descending order
    - SELECT * FROM students ORDER BY age DESC;
- Sort by multiple columns
  - SELECT * FROM students ORDER BY age DESC, name ASC;
- Sort by column position (not recommended for readability)
  - SELECT name, age FROM students ORDER BY 2;
- PostgreSQL supports NULLS FIRST / NULLS LAST to control NULL placement in sort order
  - SELECT * FROM students ORDER BY email NULLS LAST;

Operators

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Comparison Operators: =, <>/!=, <, >, <=, >=
- Logical Operators: AND, OR, NOT — combine multiple conditions
- BETWEEN – checks if a value falls within a range (inclusive)
    - SELECT * FROM students WHERE age BETWEEN 20 AND 25;
- IN – checks if a value matches any value in a list
    - SELECT * FROM students WHERE age IN (20, 22, 24);
- LIKE – pattern matching for strings
    - % – matches any sequence of characters
    - _ – matches a single character
    - SELECT * FROM students WHERE name LIKE 'A%';
- ILIKE – case-insensitive pattern matching (PostgreSQL-specific)
    - SELECT * FROM students WHERE name ILIKE 'alice';

Lecture Summary

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- SQL follows specific syntax rules: semicolons, case conventions, comments, quoted strings
- DDL commands (CREATE, ALTER, DROP, TRUNCATE) define and modify database structure
- psql meta-commands (\l, \dt, \d, \du) help inspect the database from the command line
- PostgreSQL supports rich data types: numeric, string, date/time, boolean, JSON, UUID, arrays
- DML commands (INSERT, UPDATE, DELETE) manage data within tables — always pair UPDATE/DELETE with WHERE
- SELECT retrieves data, with support for aliasing, DISTINCT, and LIMIT
- WHERE filters rows using comparison, logical, BETWEEN, IN, and LIKE/ILIKE operators
- ORDER BY sorts results, with PostgreSQL-specific NULLS FIRST/NULLS LAST control

Hands-on

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Scenario: Build an employees table for a company database
- Create company_db, connect to it (\l, \c)
- Create employees table (id, name, age, email, salary, department, hire_date)
- Inspect table (\dt, \d employees)
- Add phone column (ALTER)
- Insert 8 records (INSERT + RETURNING)
- Update salaries (UPDATE); delete one record (DELETE)
- Query with SELECT, aliasing, DISTINCT, LIMIT
- Filter with WHERE, AND, IS NULL
- Sort with ORDER BY (multi-column)
- Use BETWEEN, IN, LIKE, ILIKE

Next Lecture

- SQL Fundamentals II – Joins (inner, outer, self), subqueries, set operations